MSX and SPECTRAVIDEO

COMPUTER FORUM

Volume 1 No. 12

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MSX and SPECTRAVIDEO

COMPUTER FORUM

Magazine

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EDITORIAL

Many things have (and haven't) happened in the twelve months since Computer Forum was first published in April 1986. Had MSX really taken off in that time, Computer Forum was destined to be one of the largest-selling computer magazines in Australia.

As we all know, MSX did not have the market exposure that it deserved and the Spectravideo was discontinued. Now, after twelve months of kicking, screaming, pleading and begging with suppliers and distributors, yours truly feels just a little jaded.

The important thing is that this magazine not only survives but flourishes. Therefore, I have decided to hand over the reins to Ross Hamilton, your new editor. Ross and his team bring to Computer Forum a varied background of computer interests and expertise. Plans are already in place for a re-vitalised content and the new blood will certainly add to the enthusiasm - an essential part of producing Computer Forum.

Thank you for your support and faith in Computer Forum and I am sure that you will all give that same support to Ross and his team.

Happy computing ...

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ONE YEAR ON . . .

The past twelve months have seen some changes in the branch of computing in which we are all involved. Some computers have disappeared from the market, others have come and gone, some are still with us and we have been told of things to come.

Twelve months ago people were marketing SV318 and 328 computers and MSX computers with the brand names Sony, Mitsubishi, Pioneer, Sanyo, Spectravideo, Toshiba and Yamaha all with full support. Today, only the Yamaha is provided with any real support by its distributor.

There are the sad stories of Spectravideo SV318/328 computers being discontinued along with that company's interest in MSX. The Laser MSX which never flew. The Toshiba MSX no longer being stocked in Australia. The Mitsubishi MSX which appeared to suffer from being handled by the distributors of the Amstrad. The Sony MSX which, for a while, was market leader - heavily discounted in some locations and not others. The Sanyo MSX which, although still available, seemed to be the victim of the hot potato act within that company.

PIONEER MSX

The Pioneer PX-7 MSX slipped into the marketplace almost unnoticed. This machine was originally available for around $1000 - a specialised 32K MSX computer designed to interface with video and sound equipment. Its price tag initially made it only suitable for the Video enthusiast. However, a few of these machines were obtained at a very discounted price and we managed to get our hands on one.



The Pioneer PX-7

The PX-7 can be used with a video recorder to superimpose text or graphics over a pre-recorded video tape - great for dubbing and titling home videos. It also has the ability to control a video laser disk such as the Pioneer LD-700 or LD-1100. This is made possible by the P-Basic ROM chip which adds 16 extra commands to the MSX Basic.

The P-Basic is very easy to use as we discovered within a few minutes of unpacking the unit. At that particular time, the cricket was on TV and using the P-Basic, we were able to place an arrow on the screen and move it around in much the same manner as the TV commentators can. Another trick we tried was blanking the screen except for a circle and moved the circle around similar to the very familiar introduction to 007 movies. Unfortunately Alan Border doesn't really make a very good James Bond. Sorry Alan!

The PX-7 is a two piece unit with the keyboard being separate from the main computer unit which has built-in stereo speakers with a mixer control.

All up, the Pioneer PX-7 is a very nice unit and it is a pity they are no longer available. Those people who have them are very lucky.

THE CURRENT POSITION

Sales of SV and MSX may have seen better days but thanks to some enterprising people software is still available. There is still a large enough user base that has not outgrown the facilities provided by these machines. The immediate need is for owners to support each other and endeavour to gain satisfaction from knowing more about computers in general and SV and MSX in particular.

Nothing is to be gained from one-upmanship. You can be smug about getting something which is not available to other users but, if you are the only person in the country with a certain piece of equipment, why should anyone stock software for that equipment? Here is something for the melting pot - an SV & MSX buying group which can, with sufficient support, negotiate to obtain hardware which we are currently being denied. I can't see a company faced with an order for, say 100 disk drives, not doing all they can to supply them!

PRODUCT NEWS ...

CX5 VOICE UTILITY

CX5M owners who use the Yamaha SFG01 and SFG05 synthesisers will no doubt have encountered the problem of moving voices from one voice file to another. This task is very time consuming, if not impractical.

SOUNDWRITE have developed a utility which enables you to build new voice files from the voice files of either the SFG01 or SFG05 synthesisers. Two versions are available - one for disk and one for cassette. The utilities come on MSX cartridges and are instantly available at power on. Both versions operate in a similar manner.

After booting up and loading a voice file, the voices are displayed and you are asked to select a voice number. The utility takes a half second to move the voices and a built in type ahead buffer allows fast typists to choose voices faster than that. The voice utility also displays how many voices in your new file.

The utilities have been designed in the simplest and most robust manner possible giving protection from bad input such as voice numbers too big or unrecognised characters. Voice files created from the utilities can be loaded into either the SFG01 or SFG05 synthesisers or back to the utilities again.

So now you can set up master voice files containing all of your different voices: Keyboard, Brass, String, Percussion or whatever. When you are deciding which voices to use you can have all of any type available at once, check out which you want to use and create a new file using those voices - get the idea?

The CX5 Voice Utility is only available from Soundwrite. See their advertisement in this issue.

VIDEO TITLER

We have also been advised of a Video Titler package being developed by Soundwrite. It has Titling, Scrolling credits, Subtitles and a Leader creator. More details when it is finished in about 3 - 4 weeks.

VERSATILE MSX MEMORY BOARD

A universal printed circuit board for MSX cartridges has been developed in Australia. This board can be used for 32K or 64K RAM cartridges, EPROMS and combination EPROM and RAM. When used as a RAM cartridge, the addition of four resistors and two diodes allows battery backup to be added. The board is professionally made and available as a fully built and tested 32K RAM board installed in a cartridge ready to go - or as a blank board. See the advertisement in this issue for more details.

32K MSX RAM CARTRIDGES

Fully built and tested $80 plus $2.00 P&P anywhere in Australia or New Zealand. Blank boards suitable for RAM or EPROM or both $20 plus $1.50 postage.

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THE CX5 VOICE UTILITY

The CX5 VOICE UTILITY enables the selection of voices from a series of voice files to create a new VOICE file.

The CX5 VOICE UTILITY comes in two versions. Disk based or Cassette based. The Utility is on a Cartridge and accepts VOICE files from either the SFG01 or SFG05 synthesisers or the Utility itself.

If you want to organise your voices and have them all available for comparison with each other and rapidly create new VOICE files then this UTILITY will enable you to do it.

SEND a Bank cheque or Money Order for $65 to :-

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BUNDOORA 3083
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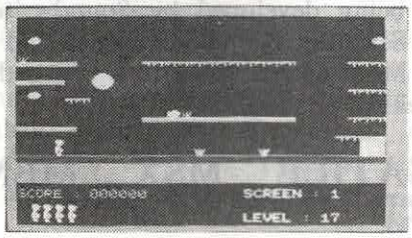
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Software NEWS & REVIEWS

GAME PACK 21 (MSX & SV Cassette) by FDS Software

This tape has two titles - GOLDRUSH and PYRAMID.

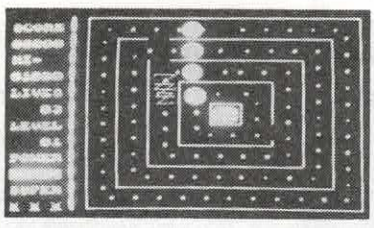


Goldrush - This game requires you as Mo the Miner, to move around the ledges of the mine collecting gold nuggets. Some of the ledges are dangerous and crumble under your weight and there are explosive charges which you might stumble into. Has seven screens and Hall of Fame scoring. Very challenging!

Pyramid - "The legends spoke of the famed 'Eye of the Gods', the huge gemstone that was said to have untold power. Now you have stumbled across an old native who knows the location of the pyramid". This is a text adventure in the classic style. Command your explorer with a verb/noun combination. Has the facility to save and load partially played games. Very difficult!

Publisher: Forrest Data Services
Recommended price: $15.00

SMACK WACKER (64K MSX Cassette)



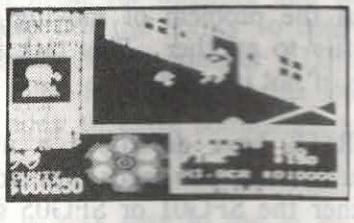
The screen is divided into tracks and you must move the teeth around these tracks collecting flourine tablets and avoiding the bacteria. You can jump from track to track as the gaps appear. Not as easy as it sounds!

Very smooth graphics.

Publisher: Aackosoft
Price: $14.95

Review copy supplied by LPG Software

MASTER GAMES (4 GAME PACK)

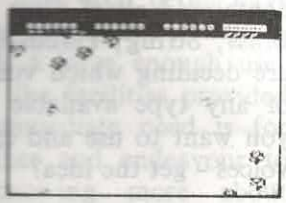


This MSX pack contains GUNFRIGHT, ALIEN 8, KNIGHTSHADE and KNIGHTLORE - all very good titles. In Gunfright, for example, you are Sheriff Quickdraw and must patrol the town looking for the baddies. The male population of the town will assist by jumping up and down and pointing in the direction of the bad guys. The women are not as helpful. In fact they're a pain. If you bump into them or they bump you then it is back to square one. If you are on horseback you can run tem over without hassle. Excellent graphics and sound.

Publisher: Ultimate
Price: $44.95

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TURBOAT (SV318/328 Cassette)


TURBOAT SD242T

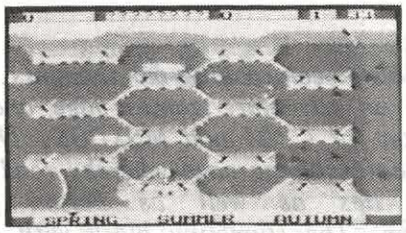
The name of the game is quick reflexes in this fast action adventure. You are the Turboat commander and your mission is to infiltrate enemy territory to pick up their secret radar for analysis by scientists.
You begin by fighting your way through the enemy's perimeter and proceed to collect the radar whilst avoiding enemy fire. Then you must escape! Simple? Wait till you try it!

1 or 2 players, good graphics and sound effects. Good concept, 1 progressive screen, 2 level

Requires extra memory for use with SV318

Publisher: Spectravideo
Price: $19.00

FIVE STAR GAMES (MSX 5 GAME PACK)

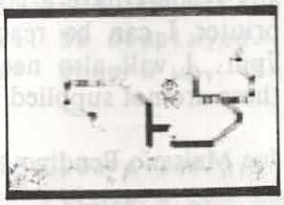


This pack contains ELIDON, MANIC MINER, BARNSTORMER, SHARKHUNTER and ZOIDS. Having already played MM and Barnstormer, I opted to review Sharkhunter. In this game you are an Eskimo fish farmer and must protect your nets which are under attack from sharks on one side and ice floes on the other. You can spear the sharks and repair the nets while swimming (but watch out for sharks). If you get eaten, the screen clears and we see poor Eskimo Nell with baby having to leave the igloo and depart. Good music, smooth graphics - I liked it.

Price: $44.95

Review copy supplied by LPG Software

BUSY BEE (SV318/328 Cassette)



BUSY BEE SD226T

This program allows you to create masterpieces on your computer screen. You control Busy Bee, who buzzes around the screen, drawing for you. You can choose from 42 different designs of line and from 10 different colours! Look out Michelangelo!

1 player, good music - sketch pad (artist) type.

Publisher: Spectravideo
Price: $12.00

MSX MSX MSX MSX
MSX MSX MSX MSX

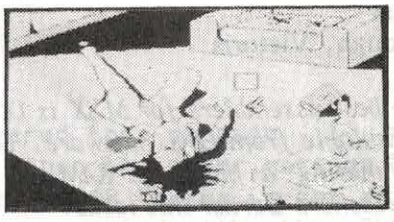
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VERA CRUZ (MSX Cassette)



This one is unusual to say the least. A beautiful mademoiselle is found dead in her apartment. You are employed by the French police to investigate her death. You must search the scene of the crime and then, using all the means at your disposal, evaluate your information and decide: murder or suicide? I won't tell you any more except that if you ever wanted to play a serious version of Inspector Clouseau - give this a try. Great concept and excellent graphics.

Publisher: Infogrames
Price: $39.95

Review copy supplied by LPG Software

MAILBOX



This page is for your comments and queries. If you have a specific problem or have solved one, we would like to hear about it.

If you require an answer before we publish it, please enclose a stamped self-addressed envelope.

ASSEMBLER HELP

I've looked everywhere for an MSX assembler and not found one. I'm very interested in computer programming and would like to make it my career. Could you please write a small machine language routine that will do the same as the following on MSX Screen 1:

```
10 X$ = HEX$ (PEEK(32775)) + HEX$ (PEEK(32774)) : X = VAL ("&B" +X$): ' Get 16-bit number
20 FOR I=6144 TO 6911 : VPOKE I,PEEK(X) : NEXT I
```

This is so I can scroll the screen in Basic.

Alex Wilson
Greensborough, Victoria

One of the best assemblers for MSX is DEVPAC 80 which is available from LPG Software. Your query regarding moving a block of RAM to VRAM is covered in Hackers Corner this issue.

DISK DRIVE HELP

I was thinking of a column you should include in your later issues. A column for people who play adventure programs - like Zork I, II, III and Hitch Hikers Guide to the Galaxy or even help for games. Please think about it - also a CP/M column for 728 and 328 users. CP/M isn't exactly easy to follow from the manuals, so information can be shared with readers. After receiving 12 copies of the magazine do I have to renew my subscription? Whatever happened to the DOS and the 100 screen game that was in Hackers Corner?

Steven Dimitrio
Footscray, Victoria

You must have been reading our thoughts. We have plans for an adventure games page and helpful information for CP/M and MSXDOS. This would be an ideas exchange page whereby readers could send in tips as well as Forum providing information. The stumbling block with adventure games is the cost of buying copies of all the available games compared to the number of readers requiring the service. Feedback please readers!!!

We would certainly like you to renew your subscription after having received your twelve issues - details are enclosed with the final issue of each reader's subscription. It seems that only you and one other person were interested in The DOS and 100 screen game. I am sure that if more readers expressed an interest, Robert Brinkworth could be persuaded to continue with these projects.

Thanks for the hints Steven which will appear in Hints and Tips

PRINTER HELP

I own an SV328 and have recently decided to purchase a dot-matrix printer. I think that an Epson would be the best choice but if you could suggest a better one it would be appreciated. If any readers have a good condition 328 compatible printer I can be reached at (054) 484 778 after 7pm. I will also need an interface and cable if these are not supplied with the printer.

Joe Maisano, Bendigo, Victoria

The Epson printer is entirely satisfactory for use with the SV328. A more economical alternative would be one of the printers reviewed last month - either is suitable for a 328. Most printers do not come with a cable, and you will require either an SV605 + cable or an SV601 or SV602 expander + printer interface + cable. Forrest Data Services manufacture a printer interface (SP101) which plugs into the back of the 328 keyboard and accepts a standard printer cable. Thank you for the hints which are published in the Hints and Tips section.

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BASIC TUTORIAL - Part 12

TURN PROMPT LINE ON/OFF

On the Spectravideo SV318/328 the bottom screen line, which displays the value of the function keys, can be turned on or off from within the SCREEN command. It was previously shown (Vol 1 No 3) that the SCREEN command was followed by two values - screen mode and sprite size. In text mode, SCREEN 0, sprite size value turns the prompt line on or off.

SCREEN 0,0 turns the prompt line off
SCREEN 0,1 turns the prompt line on

To perform the same function on SCREEN 0 with MSX computers, the KEY ON/OFF command is used -

KEY ON displays the bottom line
KEY OFF hides the bottom line

The CLEAR command

This command is used to clear a section of memory which is to contain string variables. It takes the form -

CLEAR XXXX

where XXXX is the number of bytes to be cleared for storage. If not enough space is cleared for string storage, then the error message OUT OF STRING SPACE will be displayed. An amount of 200 bytes is set aside when the computer powers up, and this can be varied with the CLEAR command.

Examples CLEAR 200 Clears 200 bytes of memory
CLEAR 0 recovers the memory set aside for string variables.
CLEAR 500 Clears 500 bytes of memory

The CLEAR command can also be used to set the TOP OF MEMORY location - the point beyond which a Basic program cannot go. The purpose being to set aside an area of memory for use by machine code routines which are accessed from Basic. To use CLEAR in this way it takes the form -

CLEAR 200,40000

This will clear 200 bytes of string storage and set top of memory at location 40000.

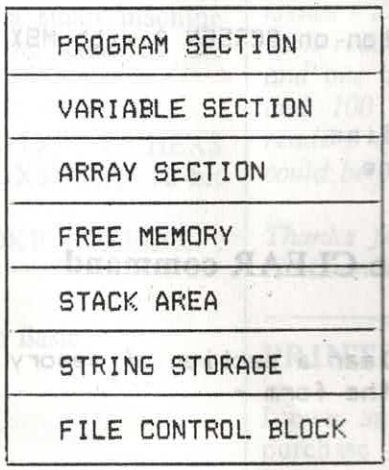
MEMORY MAP

	SV318	SV328	MSX
BASIC ROM	&H0000 - &H7FFF	&H0000 - &H7FFF	&H0000 - &H7FFF
RAM	&HC000 - &HFFFF	&H8000 - &HFFFF	&H8000 - &HFFFF

The RAM area is further broken up into two areas -

USER RAM	&HC000 - &HF4FF	&H8000 - &HF4FF	&H8000 - &HF37F
SYSTEM RAM	&HF500 - &HFFFF	&HF500 - &HFFFF	&HF380 - &HFFFF

The user RAM area is divided into sections. Apart from the Program section, they do not have a fixed start location. They expand and move according to their size and the size of the other sections.



PROGRAM SECTION: This area of RAM stores the program text as it is typed in or loaded.

VARIABLE SECTION: The size and location of string variables is stored in this area.

ARRAY SECTION: This area contains information about subscripted or array variables.

FREE MEMORY: This is the section that is left after all other sections have been allocated. Use the FRE(0) command to find its size.

STACK AREA: FOR/NEXT and GOSUBS use this area to store return addresses.

STRING STORAGE: This area stores the actual characters used for string variables and is initially set at 200 bytes after switch-on. If this is not sufficient area, then an 'Out of string space' error message will be displayed. Its size can be altered using the CLEAR command.

FILE CONTROL BLOCK (FCB): This is the area used by files and its size can be altered using the MAXFILES command.

These sections are arranged in memory in the following manner -

The PROGRAM SECTION begins at the lower limit of usable memory (&H8000 for MSX and SV328, &HC000 for SV318) and extends upward.
The VARIABLE SECTION commences immediately after the program section and extends upward.
The ARRAY SECTION extends upward from the top limit of the variable section.
The FILE CONTROL BLOCK (FCB) commences at &HF37F (MSX) or &HF4FF (SV)and extends downward toward the array section.
The STRING STORAGE area extends down from the lower limit of the FCB.
The STACK AREA commences below the string storage area and extends downward.
The FREE MEMORY area is that section of RAM which is left between the upper limit of the array section and the lower limit of the stack.

ERROR HANDLING COMMANDS

When you write a program, the normal procedure is to de-bug it until all error messages are eliminated. This is fine if you only use the program yourself, because you know what the program does and what it requires from you to perform that function. If the program is to be used by someone else or even infrequently by yourself, the situation can arise where an unforeseen error can occur and will wreck your beaut input screen display with, say, an 'extra ignored' or 'type mismatch' error message right where you don't want it. This is where you need 'error-trapping' routines.

The ON ERROR GOTO command

This command is used to move execution to the error-trap instead of crashing to the inbuilt error messages. It takes the form -

```
ON ERROR GOTO XXX
```

where XXX is the first line number of the error handling routine.

The ERR command

This is the ERRor number command. It gives the number of the last error which occured. This can be used in your error trap to determine the type of error and display your own error message and instructions about correcting the fault. For example -

```
10 ON ERROR GOTO 100
20 INPUT "NUMBERS";N
```

```
30 PRINT 20/N
40 GOTO 20
100 IF ERR=11 THEN CLS:PRINT"NOT ZERO PLEASE!":RESUME 40
110 END
```

See error codes at the end of this section.

The RESUME command

This command is used within your error-trap in a similar manner to RETURN in a subroutine. It takes three forms -

RESUME or RESUME 0 - resume program at the command generating the error.
RESUME NEXT - resume program following the command generating the error.
RESUME X - where X is the line number at which to resume the program.

The ERL command

This is the ERROR LINE command. It gives the line number in which the error occured. For example -

```
10 ON ERROR GOTO 100
20 INPUT "NUMBERS";N
30 PRINT 20/N
40 GOTO 20
100 IF ERL<>30 THEN 120
110 IF ERR=11 THEN CLS:PRINT"NOT ZERO - DUMMY!":RESUME 40
120 END
```

This version of a previous example checks to see if the error has occured in line 30 before it proceeds with the rest of the error trap.

The ERROR command

This command is used to simulate the occurence of an error. It is useful for testing your error handling routines. For example -

```
10 ON ERROR GOTO 100
15 ERROR 11
20 INPUT "NUMBERS";N
30 PRINT 20/N
40 GOTO 20
100 IF ERL=15 THEN PRINT "ERROR IN ";ERL
110 IF ERR=11 THEN CLS:PRINT"NOT ZERO - DUMMY!":RESUME 40
120 END
```

This program will cause a 'divide by zero' error to occur in line 15 and processing should jump straight to the error-trap at line 100.

ERROR CODES

CODE SV	CODE MSX	MESSAGE	MEANING
1	1	Next without For	A NEXT command has been found before a matching FOR command.
2	2	Syntax error	Cannot understand the command.
3	3	Return without Gosub	A RETURN was encountered before a GOSUB command.
4	4	Out of Data	Too many read statements for the amount of data supplied.
5	5	Illegal function call	Cannot perform the command. Requires further information.
6	6	Overflow	Result of calculations exceeds allowable value or memory location greater than 65535
7	7	Out of memory	Insufficient memory space for the function required.
8	8	Undefined line number	An attempt was made to GOTO or GOSUB a non-existant line number.
9	9	Subscript out of range	The subscripted variable exceeds the array size. A$(25) when A$ was DIM'd as 10
10	10	Redimensioned array	Attempt to change the size of an array without ERASEing it.
11	11	Division by zero	An attempt was made to divide by zero.
12	12	Illegal direct	Use of INPUT as a direct command
13	13	Type mismatch	Trying to perform string manipulation on a numeric variable and vice versa.
14	14	Out of string space	Insufficient string space was CLEARed for string functions.
15	15	String too long	String exceeds 255 characters.
16	16	String formula too complex	BASIC can't work out what you mean.
17	17	Can't continue	Can't CONTinue after a STOP or BREAK is encountered.
18	18	Undefined user function	Tried to execute a USR call before a DEFUSR statement for that routine.
19	19	Device I/O error	Interrupt to an input or output function. eg. Disk, tape.
20	20	Verify error	Error while using the CSAVE? (CSAVE verify) command.
21	21	No resume	The end of the program has been reached in error trapping mode.
22	22	Resume without error	RESUME was encountered before executing an ON ERROR GOTO
23	23	Unprintable error	Attempt to simulate an ERROR with an invalid code
24	24	Missing operand	Information given is incomplete.
25	25	Line buffer overflow	Line entered is too long.

26-49		Unprintable error	Future expansion
50	50	Field overflow	A random-access buffer was allocated more than 255 bytes.
51	51	Internal error	Disk I/O fault or error in disk operating system
52	52	Bad file number	File buffer number has not been assigned to a file using an OPEN statement.
53	53	File not found	File named is not on the disk. Check file name and extension.
54	54	File already open	Attempt to open a file that has not been closed.
55	55	Input past end	INPUT statement encountered after all information was read from a sequential file.
56	56	Bad file name	Name given to file was not of correct format.
57	57	Direct statement in file	Attempt to LOAD, RUN or MERGE a non-BASIC file.
58		Sequential after put	
59	58	Sequential I/O only	Using PUT or GET on a file which is not random access.
60	59	File not open	Attempt to input from a closed file.
61	60	Bad allocation table	Allocation table on disk does not match the one in memory.
	61	Bad file mode	Incorrect use of INPUT and OUTPUT with the OPEN command.
62	62	Bad drive number/name	
63	63	Bad track/sector	Track/sector number is incorrect
64	64	File still open	Attempt to OPEN a file which has not been closed.
65		Disk not mounted	Disk not in drive.
66		Deleted record	
67	65	File already exists	
68	66	Disk full	
	67	Too many files	Too many entries in directory.
	68	Disk write protected	
69		File write protected	Attempting to write to a file which is write protected.
70	69	Disk I/O error	Error occured while accessing a disk.
71	70	Disk offline	Disk not in drive.
72	71	Rename across disk	RENAME command cannot be used from one disk to another.

HIGHER BASIC COMMANDS

The following commands are mainly used by machine code programmers and advanced Basic programmers and are only given here for completeness.

INPUT AND OUTPUT

There are two commands used to read and write to the Input/Output (I/O) ports - INP and OUT. The I/O ports control such things as the VDP, keyboard, joysticks, disk drive and cassette.

The INP command

This command is used to check the current value being presented at an input/output (I/O) port. It takes the form -

```
A=INP(X)
```

where A is the variable to receive the value and X (0-255) is the port being tested.

This command must be used with caution because it can 'hang' or lock the computer if you do not know what you are doing.

The OUT command

This is the reverse of the INP command and is used to send data to one of the input/output (I/O) ports. It takes the form -

```
OUT X,A
```

where X is the number of the port (0-255) and A is the value (0-255) being sent to that port.

This command must be used with caution because it can 'hang' or lock the computer if you do not know what you are doing.

The DEF USR command

This command is used to define a memory address (entry point) to branch to from Basic and run a machine code routine.

The USR command

This command is used to jump to the entry point of a machine code program defined with the DEF USR command. It takes the form-

```
X=USR(N)
```

X is a 'dummy' variable and N is the value (if any) to be given to the

machine code routine from Basic.

Example

```
10 REM ENTRY POINT TO MACHINE CODE
20 DEFUSR=&HD000
30 REM JUMP TO MACHINE CODE ROUTINE
40 J=USR(0)
```

The VARPTR command

This is the variable pointer command and is used to find where variables are stored in memory and can also be used to locate a File Control Block (FCB). It takes the forms -

```
A=VARPTR(X$)
```

where A will receive the address for the variable X$.

```
A=VARPTR(#N)
```

where A will receive the address for the FCB of file number N.

SUNDRY COMMANDS

There are some commands and functions which are only used by MSX computers and some that only the Spectravideo will accept.

THE FOLLOWING COMMANDS ARE ONLY FOR USE ON SPECTRAVIDEO COMPUTERS.

The CLICK ON/OFF command

This command is used to turn the keyboard click on or off.

```
CLICK ON enables key click
CLICK OFF disables key click
```

The MSX controls the keyboard click from the SCREEN command.

The GET graphics command

This command allows you to 'grab' part of a graphics screen and store it in an array. It takes the form -

```
GET (A,B)-(C,D),N
```

where A and B are the top left co-ordinates of the area and C and D are the bottom right co-ordinates. N is the name of the array to hold the information.

The PUT graphics command

This command is used to place on the screen the graphics data which has been placed in an array with the GET command. It takes the form -

```
PUT (A,B),N
```

where A and B are the top left co-ordinates of the display location and N is the name of the array containing the data.

Example

```
10 DIM A(20,20)
20 SCREEN 1
30 DRAW"BM10,10D15R4U10R4D12R4U20"
40 GET (9,9)-(29,29),A
50 PUT (40,40),A
60 PUT (80,80),A
70 PUT (120,120),A
80 GOTO 80
```

The SWITCH command

This command is used to control another bank of RAM existing on an expansion cartridge. It cannot be used to switch in the lower 32K of an SV328. It can be used in two ways -

SWITCH will switch into the other bank
PRINT SWITCH will give 0 if the keyboard bank is being used or -1 if the RAM expansion is being used.

THE FOLLOWING COMMANDS ARE ONLY FOR USE ON MSX COMPUTERS.

The BASE command

This command is used to discover the base address of the various blocks used in the VRAM under different screen modes. It takes the form -

```
A=BASE(N)
```

where A will receive the VRAM address of base number N

	MSX SCREEN NUMBER			
	SCREEN0	SCREEN1	SCREEN2	SCREEN3
NAME TABLE	BASE(0)	BASE(5)	BASE(10)	BASE(15)
COLOR TABLE	-	BASE(6)	BASE(11)	-
PATTERN TABLE	BASE(2)	BASE(7)	BASE(12)	BASE(17)
SPRITE ATTRIBUTES	-	BASE(8)	BASE(13)	BASE(18)
SPRITE PATTERN	-	BASE(9)	BASE(14)	BASE(19)

The CALL command

This command is used to CALL a program which resides in a ROM cartridge in one of the expansion slots. The most common use for this command is CALL SYSTEM to jump to MSXDOS from Basic (if it was entered via MSXDOS). The CALL command can be replaced by the underline character.

The VDP command

This command is used to write to and read from the Video Display Processor (VDP) registers. It takes the forms -

```
VDP(X)=N
```

where X is the register number 0 to 7 (these registers are write only) and N is the value to be placed in that register.

```
N=VDP(8)
```

where N will be given the value currently stored in register 8 which is a read only register.

CONVERTING MSX PROGRAMS TO SV AND BACK

We continue this month with further hints on program conversions.

CONVERTING MSX TO SV

Printing text on a graphics screen is another task which is treated differently in both types of computers. The Spectravideo uses the LOCATE and PRINT commands just as if the graphics screen was a text screen. MSX computers cannot do this.

To print text to a graphics screen, the MSX computer requires that you OPEN a graphics (GRP:) file buffer and PRINT to it.

For example -

MSX

```
10 SCREEN 2
20 OPEN "GRP:" AS #1
30 PRINT#1,"ABC"
```

SV equivalent

```
10 SCREEN 1
20 PRINT "ABC"
```

To position the text on screen the MSX uses one of the 'Graphics Cursor' locating commands. These are PRESET, PSET and DRAW

Example 1

MSX

```
10 SCREEN 2
20 OPEN "GRP:" AS #1
30 PRESET(20,10)
40 PRINT#1,"ABC"
```

SV equivalent

```
10 SCREEN 1
20 LOCATE 20,10
30 PRINT "ABC"
```

Example 2

MSX

```
10 SCREEN 2
20 OPEN "GRP:" AS #1
30 PSET(100,50)
40 PRINT#1,"ABC"
```

SV equivalent

```
10 SCREEN 1
20 LOCATE 100,50
30 PRINT "ABC"
```

Example 3

MSX

```
10 SCREEN 2
20 OPEN "GRP:" AS #1
30 DRAW "BM128,96"
40 PRINT#1,"ABC"
```

SV equivalent

```
10 SCREEN 1
20 LOCATE 128,96
30 PRINT "ABC"
```

Printing text to a graphics screen is also subject to another very troublesome difference between these computers. The graphics screens can print a different number of characters on the screen.

MSX	SPECTRAVIDEO
SCREEN 2	SCREEN 1
24 lines	25 lines
32 characters per line	43 characters per line
SCREEN 3	SCREEN 2
6 lines	6 lines
8 characters per line	11 characters per line

NOTE: Some errors crept into PART 10 of the tutorial dealing with PRINT USING. This was theresult of using block moves on a word processor. Pages 9 to 10 include program examples which use the same line number twice. For example -

```
10 A=1234:B=76                    10 A=12.346:B=-7.5
20 F$="####"                      20 F$="+##.##"
30 PRINT USING F$;A               20 PRINT USING F$;A
30 PRINT USING F$;B               30 PRINT USING F$;B
```

Of course, in the first example the second line 30 should be 40 and in the second example the second line 20 should be 30 and 30 should be 40. Also, in the first example on page 10, line 20 should have been -

```
20 F$="**$##.##"
```

We apologise for this misleading information.

ANSWERS to the questions in issue 11

1. The valid widths for Spectravideo are 39 or 40 only.

2. NO. The only SCREEN 0 will remove sprites from the MSX screen.

3. NO. The program uses SCREEN 1 which has no direct SV equivalent.

4. MSX. To open an SV random access file would be -

```
OPEN "1:NEWFIL" AS #1 or OPEN "2:NEWFIL" AS #1
```

5. ERASE to erase the old array and its dimensions.

QUESTIONS

1. How is the prompt line turned off?

   (a) MSX
   (b) SV318/328

2. What error message will appear if not enough space is cleared using the CLEAR command?

3. Is this an MSX or SV318/328 command?

```
PUT (40,40),A
```

4. Is this an MSX or SV318/328 command?

```
PRINT BASE(0)
```

5. What is wrong with using the command SCREEN 3 with an SV318?

THE BASIC TUTORIAL CONCLUDES IN THE NEXT ISSUE

HACKERS CORNER

ACCESSING MSX ROM ROUTINES

The LDIRVM routine at &H005C is used to transfer a block of memory from main RAM to video RAM.

To use it, the start address of the block of main RAM should be loaded into the HL register pair, the destination address of the starting point in VRAM is placed in register pair DE and the length of the block in bytes is placed in the BC pair.

The LDIRMV routine at &H0059 is used to transfer a block of memory from VRAM to main RAM.

To use it, the start address of the block of VRAM should be loaded into the HL register pair, the destination address of the starting point in main RAM is placed in register pair DE and the length of the block in bytes is placed in the BC pair.

Both routines modify all registers and it is advisable to PUSH them onto the stack before calling the routines.

MSX ROM CALLS

MSX1 BIOS ENTRY TABLE

```
00CCH  Name:      ERAFNK
       Function:  Erase function key display
       Entry:     None
       Returns:   None
       Modifies:  All

00CFH  Name:      DSPFNK
       Function:  Display function keys
       Entry:     None
       Returns:   None
       Modifies:  All

00D2H  Name:      TOTEXT
       Function:  Force screen to text mode
       Entry:     None
       Returns:   None
       Modifies:  All

00D5H  Name:      GTSTCK
       Function:  Current joystick status
       Entry:     Joystick # in A
       Returns:   Direction in A
       Modifies:  All
```

THE SV ROM DECODED

Spectravideo SV-318/328 ROM version 1.1

```
798E  Set MDM flag
79C2  DIAL
79DC  DISK BOOT
79E7  No disk system
79ED  Set single density
79F0  Stop the drive
7A17  Send track number
7A1A  Send sector number
7A22  Read operation
7A60  Busy check
7A66  Bytes moved to RAM during IPL
7B44  MON
7B4A  LPT error
7B50  BOOT VECTOR (here from 0 at boot)
7CA7  Find start of RAM
7CBA  MAXFILES
7D47  Sign on message
7F97  Second bank initialise message
```


BUSINESS OPPORTUNITY

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```
00DEH  Name:      GTPDL
       Function:  Current paddle value
       Entry:     Paddle # in A
       Returns:   Value in A
       Modifies:  All

00E1H  Name:      TAPION
       Function:  Motoron & read header
       Entry:     None
       Returns:   Set carry if aborted
       Modifies:  All

00E4H  Name:      TAPIN
       Function:  Input from tape
       Entry:     None
       Returns:   Data in A
       Modifies:  All

00E7H  Name:      TAPIOF
       Function:  Stop reading tape
       Entry:     None
       Returns:   None
       Modifies:  None

00EAH  Name:      TAPOON
       Function:  Motoron & write header
       Entry:     Long header A=non-zero
                  Short header A=0
       Returns:   Carry set if aborted
       Modifies:  All

00EDH  Name:      TAPOUT
       Function:  Output to tape
       Entry:     Data in A
       Returns:   Carry set if aborted
       Modifies:  All

00F0H  Name:      TAPOOF
       Function:  Stops writing to tape
       Entry:     None
       Returns:   None
       Modifies:  None

00F3H  Name:      STMOTR
       Function:  Start cassette motor
       Entry:     Start A=1
                  Stop  A=0
                  Flip  A=255
       Returns:   None
       Modifies:  AF
```

COMPUTER KIDS

Hi Kids! I hope you all had a happy Easter. This month we have a couple of small music programs for you to type in. I think you will like them.

Bye for now, Wendy.

```
5 REM OLIVER COPYRIGHT 1987 WENDY BROWN
10 CLS:SCREEN 0:COLOR 15,13,13
15 LOCATE 5,5:PRINT"BE BACK SOON"
20 PLAY"L10O3GF#GCCAL5GL10AG#ADDO4DO3BL5
GL10GO4CO3GGB-L5AL10GEL5DL10DDL5G"
30 PLAY"L10GF#L10GCCAL5GL10AG#ADDO4DO3BL
5GL10GO4CO3GGB-L5AO4RDO3BO4C"
40 PLAY"RL10O3GCL20CO2BO3L10CGCCCGCGAO4C
O3BL5GL10GCCCEDDDBO4CL20CO3BL10O4CO3ABO4
L5D"
50 PLAY"O3L10GO4EDCO3BO4CO3L5GL10GABO4CO
3AO4L5DO3L10GCL20CO2BO3L10CADL20DC#L10DB
O4L5CO3BO4L3CR"
```

```
5 REM TRIO 1 COPYRIGHT 1987 WENDY BROWN
10 CLS:LOCATE17,10:PRINT"TRIO 1":COLOR 1
5,9
20 PLAY"T125O4L6EGFGEGL3FL6EGFGECL3D","T
125L3CDCO3C-O4CDL6CO3AL3C-","T125O4L3CO3
C-O4CO3GO4CO3AGG":COLOR15,4
30 PLAY"O4L6EGFGEGL3FL6EGFGDEL3C","L3CDC
O3C-O4CCO3C-O4C","L3CO3C-O4CO3GO4CO3AGO3
C":COLOR 15,11
40 PLAY"L6GGFFEFL3GFFL1E","L6EEDDCDL3ECD
L1C","L3O3GL6AC-O4CO3C-AGAO4CO3C-AL3AG":
COLOR 15,14
50 PLAY"L3O4FGGFFEFE","L6CCDEFEDCFECL3DC
","L3O3AC-O4CO3AGAL6GC-L3O4C":COLOR 15,5
60 PLAY"L6EGFGEGL3FL6EGFGECL3D","L6CEDEC
ECO3C-O4CEDEL3CO3C-","O4L3CO3C-AGAC-L6O4
CO3AL3G":COLOR 15,1
70 PLAY"O4L6EGFGEGL3FL6EGFGT100DEL3C","L
3CDEDCCT100O3C-O4C","L3O3GL6AC-O4CO3C-AG
AC-AGT100L3GO3C":COLOR 15,2:FORZ=1TO1000
:NEXTZ
80 COLOR 15,6:FOR A=1TO1000:NEXT:COLOR 1
5,10:FORB=1TO1000:NEXT:COLOR 15,4:FORF=1
TO1000:NEXT
```

UNDERSTANDING MACHINE CODE

THE Z80 INSTRUCTIONS

There are 144 instructions, or mnemonics, which can be used in nearly 700 variations. These instructions must be assembled into a program. This can be done by hand (a frustrating, error prone task) or with the use of an Assembler.

A machine code assembler allows you to write machine code programs using the instruction set without having to remember where sections are located in memory. In Basic, for example, we might have a subroutine at line 1000 which we can access with the command GOSUB 1000. The nearest equivalent instruction in machine code is CALL xxxx (xxxx is a memory location). This means that you must remember the memory location each time you wish to CALL (or GOSUB) that routine. An Assembler allows you to use a LABEL or name for the routine and simply use CALL LABEL to access it.

When using an Assembler, you enter the instruction based on the instruction set. For example, if we commenced assembling at memory location 40960, then to load the number 28 into the A register you would type

LD A,28

The Assembler would then examine what you have just typed and look up the 'machine code' equivalent of your assembly language instruction LD A and store the value 62 into the current memory location and 28 into the next memory location. This is referred to as a two-byte instruction and would appear in memory as -

Memory location	contents
40960	62
40961	28

At this point it should be mentioned that most machine code or assembly language programs use the hexadecimal system exclusively. That is, all memory addresses, instruction codes and values are expressed in their hexadecimal values. Sometimes they are preceded by &H but more normally the number will be followed by the letter H. Our little example program would be more correctly shown as -

Memory location	contents
A000H	3EH
A001H	1CH

A three-byte instruction such as LD HL,B000H which means load register pair HL with the number B000H would appear in memory as

Memory location	contents
A000H	21H
A001H	00H
A002H	B0H

Notice how the code 21H for the instruction LD HL is stored in location A000 and then the value B000H has its second part 00 stored first followed by its first part B0. This is called 'storing a 2-byte integer' where the 'low byte' or L is stored first followed by the 'high byte' or H.

EXAMPLE INSTRUCTIONS

Instruction	Code	Meaning
LD A,B	78H	Load register A with the contents of register B
CALL A000H	CDH,00H,A0H	Jump to the sub-routine which commences at mem-ory location A000H
INC A	3CH	Add one to the value in reg. A
DEC B	05H	Reduce the value in the A register by one

READERS ADS

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HINTS & TIPS

Joe Maisano of Bendigo sends in these tips which are for the SV318/328

To get the Spectravideo symbol-

DEFUSR=&H4782 : PRINT USR(&H4782)

To hang up the computer -

DEFUSR=&H210 : PRINT USR(&H210)

To reset the computer -

DEFUSR=&H625 : PRINT USR(&H625)

Steven Dimitrio of Footscray provides the following tips -

For faster disk turnoff on the Spectravideo use

POKE &HD993,50 : POKE &HD994,0

Turn on caps lock -

OUT &H88,15 : OUT &H8C,255 : POKE &HFE38,1

Place both routines at the beginning of ipl in Disk Basic to operate at power up.

PROGRAM LISTINGS

Each issue of COMPUTER FORUM will contain program listings for you to type in. When entering these programs it is a good idea to save off to cassette or disk every 10 or 20 lines and then continue. Then if the power goes off or another major disaster occurs, you will not have to type in the whole lot again.

Another check is to verify each save using the CSAVE? command. To use this command, CSAVE your program, then rewind the tape and type CLOAD? and press enter. This command will then compare the program on the tape with the one in memory, to verify that it was saved correctly.

All of the listings are printed in 40 column format. If you enter WIDTH40 before typing them in, you will be able to line up the characters in the listing with the text on screen.

Take care that you do not mistake the letter l for 1 or an O for a zero. Where graphics characters are used in the program, they will be shown as @1@. This symbol will then be repeated at the end of the listing with instructions concerning the graphics character used.

All of the listings have been tested on MSX and SV318/328 computers and do work. Should you have any problems, we suggest that you thoroughly check your work as errors can creep in very easily. Should you not wish to type in these listings, send a blank tape together with $2 to COMPUTER FORUM and we will post a copy to you. Do not forget to tell us which computer you have.

REAL MARTIANS DON'T EAT APPLES
by Michael Porter

```
This listing is in Spectravideo format.
For MSX owners, see note at the end of
the listing.

10 CLS:SCREEN2:COLOR4,7,7:CLS:LOCATE80,0
:PRINT"REAL":LOCATE30,32:PRINT"MARTIANS"
:LOCATE70,64:PRINT"DON'T":LOCATE90,96:PR
INT"EAT":LOCATE50,128:PRINT"APPLES"
15 PLAY"O4T255V15L5EDL3CR8L5EDL3CR8L5GL6
FR30L8FL3ER8L5GL6FR30L8FL3E":FOROP=1TO10
00:NEXT:R=RND(-TIME)
```

```
20 PLAY"O4T255V15L5EDL3CR8L5EDL3CR8L5GL6
FR30L8FL3ER8L5GL6FR30L8FL3E":FOROP=1TO10
00:NEXT:R=RND(-TIME)
30 DEFINTA-O
40 CLEAR1000:RESTORE
50 PLAY"O4T255V15L5EDL3CR8L5EDL3CR8L5GL6
FR30L8FL3ER8L5GL6FR30L8FL3E":N=0:SCREEN1
:COLOR15,4,4:CLS
60 LOCATE47,10:PRINT"TO SELECT GAME OPTI
ON,":LOCATE47,26:PRINT"PRESS KEY ON KEYB
OARD.":FORI=1TO8:LOCATE42,34+I*16:PRINTI
"= SKILL"I:LOCATE114,34+I*16:PRINT"/ONE
PLAYER":NEXT
70 SP$=INKEY$:IFSP$=""THEN70
80 IFSP$<"1"ORSP$>"8"THEN70
90 SP=VAL(SP$)
100 A=0:B=5:CLS:ONSPRITEGOSUB1020
110 SCREEN1,2:COLOR2,1,1:CLS:SPRITEON
120 RESTORE
130 S1$="":S2$="":FORQ=1TO16:READA1:READ
A2:S1$=S1$+CHR$(A1):S2$=S2$+CHR$(A2):NEX
T
140 S$=S1$+S2$
150 SPRITE$(0)=S$
160 FORI=0TO8:READX%(I),Y%(I):NEXT
170 Z$=STRING$(32,95)
180 W$=STRING$(32,166)
190 LOCATE0,30:PRINTZ$
200 LOCATE0,38:PRINTW$
210 LOCATE0,41:PRINTZ$
220 LOCATE0,70:PRINTZ$
230 LOCATE0,78:PRINTW$
240 LOCATE0,81:PRINTZ$
250 LOCATE0,110:PRINTZ$
260 LOCATE0,118:PRINTW$
270 LOCATE0,121:PRINTZ$
280 LOCATE0,150:PRINTZ$
290 LOCATE0,158:PRINTW$
300 LOCATE0,161:PRINTZ$
310 LOCATE0,190:PRINTZ$
320 LOCATE0,198:PRINTW$
330 LOCATE0,201:PRINTZ$
340 LOCATE0,231:PRINTW$
350 A=STICK(0):PUTSPRITE0,(128,99),7,0
360 S$=""
370 S1$="":S2$="":FORQ=1TO16:READA1:READ
A2:S1$=S1$+CHR$(A1):S2$=S2$+CHR$(A2):NEX
T
380 S$=S1$+S2$
390 SPRITE$(3)=S$
400 LINE(12,25)-(18,169),15,B:LINE(152,2
5)-(158,169),15,B
410 FORT=25TO161STEP8:LINE(12,T)-(18,T):
LINE(152,T)-(158,T):NEXT
420 PSET(192,0):DRAW"R64D192L64U192"
430 COLOR15,1,1:LOCATE200,100:PRINT"SCOR
E":LOCATE200,110:PRINTA:LOCATE200,150:PR
INT"HIGH":LOCATE200,160: COLOR 1,1:PRINT
OH:LOCATE200,160:COLOR15,1:PRINTHS
440 DATA 7,128,15,192,31,224,51,48,51,48
,51,48,63,240,56,112
450 DATA 44,208,39,144,23,160,4,128,12,1
92,12,192,12,192,28,224
460 DATA0,0,0,-5,0,0,5,0,0,0,0,5,0,0,-5,
0,0,0
470 LOCATE200,5:COLOR1,1:PRINT"LIVES"B+1
:COLOR15,1:LOCATE200,5:PRINT"LIVES"B
480 DATA 0,0,0,0,0,0,0,0,3,0,7,128,15,19
2,27,96
490 DATA 59,112,95,232,143,196,135,132,7
,128,7,128,12,192,28,224
500 DATA 0,0,0,0,0,0,0,0,3,0,7,128,15,19
2,27,96
510 DATA 59,112,95,232,47,208,23,160,7,1
28,7,128,12,192,28,224
520 S1$="":S2$="":FORQ=1TO16:READA1:READ
A2:S1$=S1$+CHR$(A1):S2$=S2$+CHR$(A2):NEX
T
530 S$=S1$+S2$
540 SPRITE$(4)=S$
550 DATA 8,32,7,128,15,192,31,224,51,48,
51,48,51,48,63,240
```

```
560 DATA 184,116,204,204,7,128,135,132,1
24,248,56,112,0,0,0,0
570 DATA 1,128,2,0,10,64,30,224,63,128,6
3,0,63,0,63,0
580 DATA 63,128,31,224,15,192,4,128,0,0,
0,0,0,0,0,0
590 S1$="":S2$="":FORQ=1TO16:READA1:READ
A2:S1$=S1$+CHR$(A1):S2$=S2$+CHR$(A2):NEX
T:FORE=1TO4:TV(E)=90:NEXT
600 S$=S1$+S2$:SPRITE$(5)=S$
610 S1$="":S2$="":FORQ=1TO16:READA1:READ
A2:S1$=S1$+CHR$(A1):S2$=S2$+CHR$(A2):NEX
T
620 S$=S1$+S2$:SPRITE$(6)=S$
630 H=INT(RND(1)*4+1):F=(H-1)*40+7:G=INT
(RND(1)*110+20)
640 PUTSPRITE7,(G,F),8,6
650 FORI%=1TO4:EX%(I%)=0:DX%(I%)=2+SP*2:
PUTSPRITE20+I%,(EX%(I%),40*I%-21),5,3:NE
XT
660 LOCATE200,5:COLOR1,1:PRINT"LIVES"B+1
:COLOR15,1:LOCATE200,5:PRINT"LIVES"B
670 SPRITEON:E=128:D=99:PUTSPRITE0,(E,D)
,7,0:I=3:C=99:RT=5
680 M=STICK(0):IFM=0THEN M=STICK(1)
690 IFM=0THENPS=PS+1:IFPS=100THENPS=0:GO
SUB1040
700 IFM=10RM=5THENGOSUB840
710 IFN=0THENE=E+X%(M)
720 IFJORSTRIG(0)ORSTRIG(1)THENGOSUB960
730 IFE<3THENE=3
740 IFE>178THENE=178
750 PUTSPRITE0,(E,D),7,0
760 EX%(I)=EX%(I)+DX%(I)
770 IFEX%(I)<3OREX%(I)>176THENDX%(I)=-DX
%(I)
780 IFO=3THENO=4:GOTO810
790 O=3
800 IFSP>4THENIFRND(1)>SP/10THENCT=(CT+1
)AND1:RT=1+CT*4
810 PUTSPRITEI+20,(EX%(I),C),RT,O
820 IFG>=170THENG=0
830 G=G+SP:PUTSPRITE7,(G,F),2,6:GOTO680
840 IFD<>FTHENSPRITEON
850 IFE=>6ANDE<9THEND=D+Y%(M)
860 IFE<149ANDE>146THEND=D+Y%(M)
870 IFD<43THENI=1:C=19
880 IFD>43ANDD<82THENI=2:C=59
890 IFD>82ANDD<120THENI=3:C=99
900 IFD>120ANDD<190THENI=4:C=139
910 IFD<14THEND=14
920 IFD>144THEND=144
930 IFD<>19ANDD<>59ANDD<>99ANDD<>139THEN
N=1:GOTO950
940 N=0
950 RETURN
960 IFD=C-15THEN1000
970 IFJ=1THENJ=0:RETURN
980 IFN=0THENJ=1:D=D-15:N=0:PLAY"V1504S7
G":E=E+X%(M):RETURN
990 RETURN
1000 SPRITEOFF:K=K+1:IFK=3THEND=C:K=0:E=
E+X%(M):J=0:PUTSPRITE0,(E,C),7,0:SPRITEO
N:RETURN
1010 RETURN
1020 IFD=C-15THEN1060
1030 IFE>EX%(I)+14ORE<EX%(I)-15THEN1060
1040 PLAY"V1501G":SPRITEOFF:N=0:J=0:B=B-
1:IFB>0THENGOTO630
1050 GOTO1110
1060 SPRITEOFF:A=A+50:N=0:PLAY"O4V15G":I
FA/500=INT(A/500)THENPLAY"V1506BAGBAG":B
=B+1:SP=SP+1:FORS=1TO4:DX%(X)=SP:NEXT
1070 LOCATE200,100:PRINT"SCORE":LOCATE20
0,110:COLOR1,1:PRINTSTRING$(10,201):COLO
R15,1:LOCATE200,110:PRINTA
1080 LOCATE200,5:COLOR1,1:PRINT"LIVES";S
TRING$(2,201):COLOR15,1:LOCATE200,5:PRIN
T"LIVES"B
1090 H=INT(RND(1)*4+1):F=(H-1)*40+7:G=IN
T(RND(1)*110+20):SPRITEOFF
1100 PUTSPRITE7,(G,F),8,6:SPRITEOFF:RETU
RN
```

```
1110 OH=HS:IFA>HSTHENHS=A
1120 SCREEN2:COLOR4,7,7:CLS:LOCATE20,10:
PRINT"GAME OVER":LOCATE60,80:PRINT"SCORE
":LOCATE50,150:PRINTA:FORI=1TO1000:NEXT:
GOTO50
```

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NOTE: MSX owners change the following lines to read –

```
10 CLS:SCREEN3:OPEN"GRP:"AS#1:COLOR4,7,7
:CLS:PRESET(80,0):PRINT#1,"REAL":PRESET(
30,32):PRINT#1,"MARTIANS":PRESET(70,64):
PRINT#1,"DON'T":PRESET(90,96):PRINT#1,"E
AT":PRESET(50,128):PRINT#1,"APPLES"
60 OPEN"GRP:"AS#1:PRESET(47,10):PRINT#1,
"TO SELECT GAME OPTION,":PRESET(47,26):P
RINT#1,"PRESS KEY ON KEYBOARD.":FORI=1TO
8:PRESET(42,34+I*16):PRINT#1,I"= SKILL"I
:PRESET(144,34+I*16):PRINT#1,"/ONE PLAYE
R":NEXT
170 Z$=STRING$(24,95)
180 W$=STRING$(24,193)
190 PRESET(0,30):PRINT#1,Z$
200 PRESET(0,38):PRINT#1,W$
210 PRESET(0,41):PRINT#1,Z$
220 PRESET(0,70):PRINT#1,Z$
230 PRESET(0,78):PRINT#1,W$
240 PRESET(0,81):PRINT#1,Z$
250 PRESET(0,110):PRINT#1,Z$
260 PRESET(0,118):PRINT#1,W$
270 PRESET(0,121):PRINT#1,Z$
280 PRESET(0,150):PRINT#1,Z$
290 PRESET(0,158):PRINT#1,W$
300 PRESET(0,161):PRINT#1,Z$
310 PRESET(0,190):PRINT#1,Z$
320 PRESET(0,198):PRINT#1,W$
330 PRESET(0,201):PRINT#1,Z$
340 PRESET(0,231):PRINT#1,W$
430 COLOR15,1,1:PRESET(200,100):PRINT#1,
"SCORE":PRESET(200,110):PRINT#1,A:PRESET
(200,150):PRINT#1,"HIGH":PRESET(200,160)
:COLOR1,1:PRINT#1,OH:PRESET(200,160):COL
OR15,1:PRINT#1,HS
1070 PRESET(200,100):PRINT#1,"SCORE":PRE
SET(200,110):COLOR1,1:PRINT#1,STRING$(6,
219):COLOR15,1:PRESET(200,110):PRINT#1,A
1080 PRESET(200,5):COLOR1,1:PRINT#1,"LIV
ES";STRING$(2,219):COLOR15,1:PRESET(200,
5):PRINT#1,"LIVES"B
1120 SCREEN3:COLOR4,7,7:CLS:PRESET(20,10
):PRINT#1,"GAME OVER":PRESET(60,80):PRIN
T#1,"SCORE":PRESET(50,150):PRINT#1,A:FOR
I=1TO1000:NEXT:GOTO50
```

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